

ArithmeticSite

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Chapter 1

The multiplicative monoid $\mathbb{N}^\times$

The Arithmetic Site, introduced by Connes and Consani, is a geometric object — a topos equipped with a structure sheaf — whose points in the topos-theoretic sense are in natural bijection with the adèle class space of $\mathbb{Q}$. Its construction begins with a single, elementary monoid: the positive integers under multiplication. Everything else — the tropical semiring, the presheaf topos, the structure sheaf, and ultimately the connection to the Riemann zeta function — is built on top of this one piece of data.

Definition 1.1. Let $\mathbb{N}^\times$ denote the set $\{1, 2, 3, \dots\}$ of strictly positive integers equipped with ordinary multiplication. This is a commutative monoid with unit 1. In Lean we represent this as `PNat` with its multiplicative structure, via the abbreviation `NPos`.

The unique factorisation theorem says that $\mathbb{N}^\times$ is the free commutative monoid on the set of primes: every element factors uniquely as a finite product $\prod_p p^{a_p}$. This factorisation structure is not merely background — it will resurface as the combinatorial backbone of the site structure on $B\mathbb{N}^\times$, where factorizations of integers into coprime parts give rise to covering families whose overlap conditions reflect divisibility.

Definition 1.2. The *classifying category* $B\mathbb{N}^\times$ of $\mathbb{N}^\times$ is the one-object category whose single object is $*$ and whose morphisms $*$ $\rightarrow$ $*$ are the elements of $\mathbb{N}^\times$, with composition given by multiplication.

Remark 1.3. The passage from a monoid M to the one-object category BM is the standard *delooping* construction. A presheaf on BM — that is, a functor $BM^{\text{op}} \rightarrow \mathbf{Set}$ — is exactly a set X equipped with a right action of M : each element $n \in M$ acts as a function $X \rightarrow X$, and the action is compatible with composition. The presheaf topos $\widehat{B\mathbb{N}^\times}$ introduced in Chapter 3 is precisely the category of such sets-with- $\mathbb{N}^\times$ -action, and it is in this topos that the Arithmetic Site lives.

Chapter 2

The tropical semiring $\bar{\mathbb{N}}$

The tropical semiring $\bar{\mathbb{N}}$ is the second fundamental ingredient of the Arithmetic Site. It plays the role of the *structure sheaf*: as Connes and Consani put it, $\bar{\mathbb{N}}$ is the semiring on which $\mathbb{N}^\times$ acts by multiplication, and it is this action that makes $\bar{\mathbb{N}}$ into a sheaf of semirings over the topos $\widehat{B\mathbb{N}^\times}$. The arithmetic of $\bar{\mathbb{N}}$ is characteristic-1 arithmetic — addition is idempotent — which is the tropical, or max-plus, shadow of ordinary arithmetic.

Definition 2.1. The *tropical semiring* $\bar{\mathbb{N}}$ is the set $\mathbb{N} \cup \{+\infty\}$ equipped with:

- *Tropical addition*: $a \oplus b := \min(a, b)$, with identity element $+\infty$.
- *Tropical multiplication*: $a \odot b := a + b$ (ordinary addition), with identity element 0.

In Lean, $\bar{\mathbb{N}}$ is represented as `Tropical (WithTop ℕ)`, via the abbreviation `NBar`. The zero element is `trop ⊤` (i.e. $+\infty$) and the multiplicative identity is `trop 0`.

Lemma 2.2. $\bar{\mathbb{N}}$ is a commutative semiring.

Proof. This is the Mathlib instance `CommSemiring (Tropical (WithTop ℕ))`, found by `inferInstance`. □

Lemma 2.3. Tropical addition on $\bar{\mathbb{N}}$ is idempotent: $a \oplus a = a$ for all $a \in \bar{\mathbb{N}}$.

Proof. $a \oplus a = \min(a, a) = a$. In Lean this follows from `Tropical.add_eq_left`. □

Remark 2.4. Idempotency of addition — equivalently, the identity $1 \oplus 1 = 1$ — is the defining feature of *characteristic 1* arithmetic. It is what places $\bar{\mathbb{N}}$ in the same conceptual family as finite fields $\mathbb{F}_p$ (characteristic p), with $\min$ playing the role that $+\text{ mod } p$ plays there. The scaling endomorphisms defined below are the characteristic-1 analogue of the Frobenius endomorphism $x \mapsto x^p$ of $\mathbb{F}_p$.

Definition 2.5. For each $n \in \mathbb{N}^\times$, the *scaling endomorphism* $\varphi_n : \bar{\mathbb{N}} \rightarrow \bar{\mathbb{N}}$ is defined by

$$\varphi_n(x) := nx \quad (x \in \mathbb{N}), \quad \varphi_n(+\infty) := +\infty,$$

where nx denotes ordinary multiplication in $\mathbb{N}$. Equivalently, $\varphi_n(x) = x^{\odot n}$, the n -fold tropical product of x with itself. In Lean, φ_n is defined as a semiring endomorphism `NBar ->+* NBar` via `scalingEndomorphism`.

Lemma 2.6. *For each $n \in \mathbb{N}^\times$, the map φ_n is a semiring endomorphism of $\bar{\mathbb{N}}$: it preserves both operations and both identities.*

Proof. Preservation of $+\infty$ and of tropical multiplication are immediate. Preservation of tropical addition uses $n \cdot \min(a, b) = \min(na, nb)$ for $n > 0$, which holds because scaling by n is order-preserving on $\mathbb{N} \cup \{+\infty\}$. In Lean this is the `map_add'` field of `scalingEndomorphism`, proved via monotonicity of `nsmul` and a case split on `le_total`. $\square$

Lemma 2.7. *The scaling endomorphisms compose multiplicatively:*

$$\varphi_m \circ \varphi_n = \varphi_{mn}$$

for all $m, n \in \mathbb{N}^\times$. In particular, the map $n \mapsto \varphi_n$ is a monoid homomorphism from $(\mathbb{N}^\times, \times)$ into the semiring endomorphism monoid $\text{End}_{\text{Semiring}}(\bar{\mathbb{N}})$.

Proof. $(\varphi_m \circ \varphi_n)(x) = m(nx) = (mn)x = \varphi_{mn}(x)$. In Lean this is `scalingEndomorphism_comp`, proved via `pow_mul`. $\square$

Remark 2.8. The Frobenius *correspondences* $\Psi(\lambda)$, parametrised by $\lambda \in \mathbb{Q}_+^\times$, are a more refined object: they live in the *square* $\widehat{B\mathbb{N}^{\times 2}}$ of the Arithmetic Site and are defined via the action of $\mathbb{N}^\times \times \mathbb{N}^\times$ on $\bar{\mathbb{N}} \otimes_{\mathbb{B}} \bar{\mathbb{N}}$. They are introduced in Chapter 7.

Chapter 3

The presheaf topos $\widehat{\mathbb{N}^\times}$

Definition 3.1. The *presheaf topos* $\widehat{\mathbb{N}^\times}$ is the category of functors $B\mathbb{N}^{\times\text{op}} \rightarrow \mathbf{Set}$, i.e. the category of sets equipped with a left $\mathbb{N}^\times$ -action. In Lean this is `Action (Type*) (Multiplicative ℕ+)`.

Lemma 3.2. $\widehat{\mathbb{N}^\times}$ is a Grothendieck topos.

Proof. Any presheaf category on a small category is a Grothendieck topos. □

Chapter 4

The structure sheaf

Definition 4.1. The *structure sheaf* $\mathcal{O}$ of the Arithmetic Site is the object of $\widehat{\mathbb{N}^\times}$ whose underlying set is $\bar{\mathbb{N}}$ and whose $\mathbb{N}^\times$ -action is given by the scaling endomorphisms: $n \cdot x := \varphi_n(x) = nx$.

Lemma 4.2. The assignment $n \mapsto \varphi_n$ defines a monoid homomorphism $(\mathbb{N}^\times, \times) \rightarrow \text{End}_{\text{Semiring}}(\bar{\mathbb{N}})$, making $\bar{\mathbb{N}}$ into a semiring object of $\widehat{\mathbb{N}^\times}$ with $\mathbb{N}^\times$ acting by multiplication.

Chapter 5

The Arithmetic Site

Definition 5.1. The *Arithmetic Site* is the pair $(\widehat{\mathbb{N}^\times}, \mathcal{O})$ consisting of the presheaf topos $\widehat{\mathbb{N}^\times}$ together with its structure sheaf $\mathcal{O}$.

Chapter 6

Points of the topos

Definition 6.1. A *point* of a topos $\mathcal{E}$ is a geometric morphism $p : \mathbf{Set} \rightarrow \mathcal{E}$, i.e. an adjoint pair $p^* \dashv p_*$ where p^* is left exact.

Theorem 6.2. *The points of $\widehat{\mathbb{N}^\times}$ are in natural bijection with the adèle class space $\mathbb{Q}_+^\times \backslash \mathbb{A}^f / \widehat{\mathbb{Z}}^*$.*

Proof. (sorry)

□

Chapter 7

The Frobenius correspondences

Definition 7.1. For each $\lambda \in \mathbb{Q}_+^\times$, the *Frobenius correspondence* Φ_λ is a correspondence on $\widehat{\mathbb{N}^\times} \times \widehat{\mathbb{N}^\times}$ arising from the scaling action of $\mathbb{Q}_+^\times$ on the adèle class space.

Theorem 7.2. *The Frobenius correspondences realize the scaling action of $\mathbb{Q}_+^\times$ on the adèle class space at the level of the topos $\widehat{\mathbb{N}^\times}$.*

Proof. (sorry)

□

Chapter 8

Connection to the Riemann zeta function

Theorem 8.1. *The Hasse–Weil zeta function of the Arithmetic Site recovers the Riemann zeta function $\zeta(s)$.*

Proof. (sorry)

□